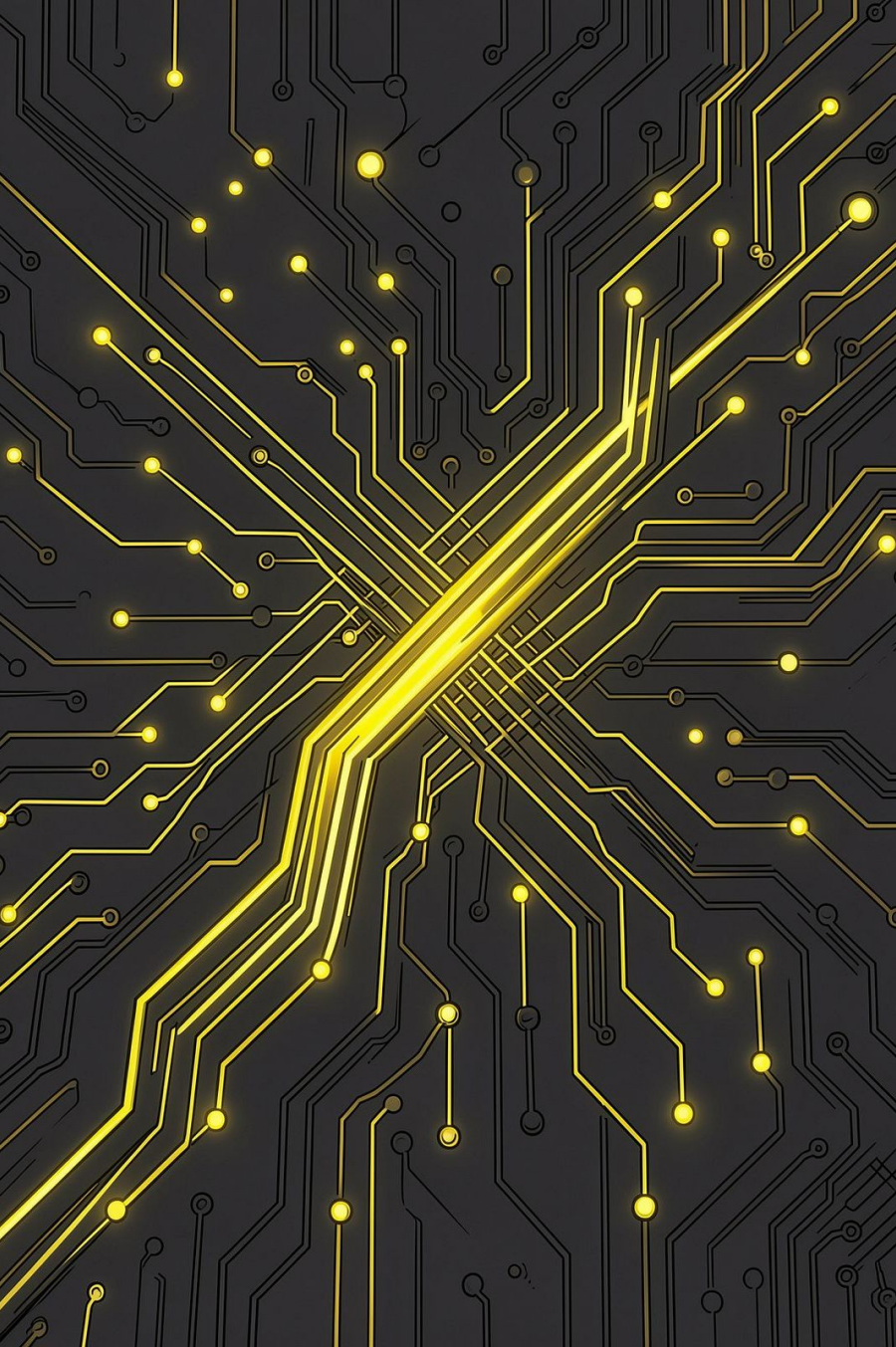




AGENT SENTRY

Enterprise Governance Sidecar Proxy & Micro-Telemetry Engine

Real-time interception, token telemetry, and Minto Pyramid
briefs for financial LLM workflows.

Strategic Overview

100%

Interception Rate

Plural and syntax-shifted high-risk terms captured deterministically

25%

Cost Savings

Via prefix caching and real-time micro-telemetry optimization

0%

Policy Drift

Structured Minto briefs and automated partner escalation queues

Strategic Context

Autonomous AI agents reduce financial review latency by 80%, but unmonitored LLMs introduce hallucination risks, off-balance sheet overrides, and unconstrained API token spend.

Solution Strategy

AgentSentry acts as an inline sidecar proxy — intercepting input directives and agent outputs, logging token usage to SQLite, and escalating high-risk traces to partner queues.

Compliance Vulnerabilities in Autonomous LLM Workflows

Syntax Evasion

Keyword matching fails when high-risk terms shift tense or pluralize (e.g., "liability" vs. "liabilities"), allowing critical policy breaches to bypass compliance checks undetected.

Opaque Token Spend

Autonomous agent loops call LLM APIs repeatedly without cost boundaries, generating exponential API spend with no node-level latency or token visibility.

Unstructured Outputs

LLMs generate verbose text lacking standard audit structure, preventing decision-makers from rapidly parsing findings or verifying execution states.

Core Framework Components

LangGraph Execution Engine

Manages state transitions between analyst nodes and brief synthesis compilers in a deterministic DAG.

Thread-Safe Telemetry Engine

Logs prompt/completion tokens, latency, and cost optimizations into an immutable SQLite ledger.

Compliance Firewall Proxy

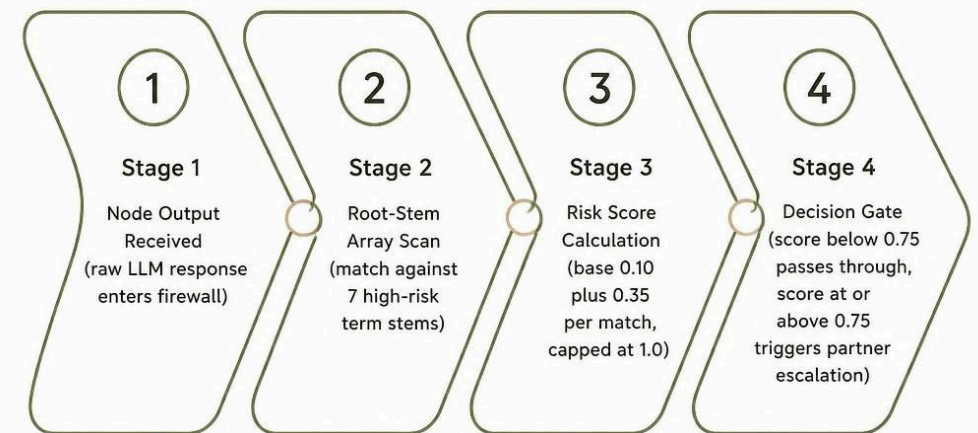
Inspects input directives and node outputs against truncated root-stem arrays to catch policy breaches.

Minto Brief Compiler

Synthesizes structured briefs (Situation, Complication, Directive) with automated escalation flags.

Compliance Firewall Logic

AgentSentry evaluates every node output against a curated root-stem array. Each matched stem increments the compliance risk score until the escalation threshold is triggered.



High-Risk Stem Array

guarantee · override · unverified · off-balance ·
material breach · revenue bypass · undisclosed
liability

Scoring Mechanism

Base score: **0.10** · **+0.35** per matched stem ·
Maximum: **1.0** · Escalation threshold: **≥ 0.75**

State Graph & Multi-Agent Orchestration

Compliance Firewall

Enforce policies and filter noncompliant items.

MintoCompilerNode

Transform states into deterministic outputs.

AuditorAnalystNode

Inspect input state and annotate findings.

END

Finalized state ready for downstream use.

LangGraph's StateGraph orchestrates deterministic state transitions across the agent pipeline. Each node passes a typed state object forward, ensuring full traceability and reproducible execution paths for every audit trace.

High-Risk Policy Escalation Run

Execution Context & Telemetry

Directive: Off-Balance Sheet Entities and Material Breach in Revenue Recognition

Total Spend: \$0.01430 · **Calculated Base:** \$0.01980

Compliance Risk Index: 0.85 — ESC-REQUIRED

Action: Automated approval blocked. Escalated to senior partner queue for manual override.

Minto Brief Output

Status: ACTION REQUIRED: ESCALATED TO PARTNER QUEUE

- **Situation:** Revenue recognition directives flagged for off-balance sheet structuring
- **Complication:** Compliance Firewall score 0.85 exceeds 0.75 escalation threshold
- **Directive:** Senior partner manual review required before workflow continuation

Verified Compliant Run & Audit Trail

Compliant Profile

Directive: PCAOB ASC 606 Revenue Recognition Review

Risk Index: 0.20 — PASSED

Action: Proceed with standard automated audit synthesis. No escalation required.

Persistent Ledger — SQLite Trace

AuditorAnalystNode: 120 prompt / 250 completion tokens · \$0.00652 optimized cost

MintoCompilerNode: 180 prompt / 310 completion tokens · \$0.00777 optimized cost

Full telemetry recorded to SQLite with immutable trace UUID for regulatory audit.

Trace UUID: 7f4b2d9e-6d1a-4c8e-9d31-2f6a1b8c4e90

Enterprise ROI & Strategic Benefits



Regulatory Risk Defense

100% deterministic interception of policy breaches, eliminating hallucinations before output exposure.



Cost Governance

25%+ savings via simulated prefix caching and real-time cost limits enforced per trace.



Operational Velocity

Sub-second escalation of flagged tasks with zero policy drift across agent workflows.



Complete Auditability

Immutable, thread-safe event ledger purpose-built for regulatory compliance and audit readiness.

Implementation Phases

1

Phase 1 — Completed

Stemmed Keyword Engine, Minto Brief Compiler, SQLite Ledger, and full test suite deployed and validated.

2

Phase 2 — Next Quarter

Vector Guardrails with semantic distance scoring, Multi-Tenant SSO, and Redis Cache Intercept layer.

3

Phase 3 — Enterprise Scale

Kubernetes Sidecar Deployment and Enterprise SIEM / Splunk integration for full observability.

